



+ + + PERFORMANCE
TESTING



+ THE GROWTH OF PERFORMANCE WEAR

In recent years, the sportswear industry has influenced mainstream fashion in a significant way. Factors such as the growing portion of the population seeking healthier lifestyles, the sustained influence of athleisure, and technological developments designed to improve comfort, means performance wear is a trend that is predicted to grow substantially over the next decade.

Compared with mainstream fashion garments, which are about aesthetic appearance or decorative characteristics; performance fabrics have a primarily technical/functional benefit such as 'comfort' (e.g. keeping the wearer warm/cool/dry by regulating the body temperature), and/or 'safety' (e.g. protecting the wearer from the environment, and hazards such as fire-proofing the outer layer of the garment). The rise in performance wear has resulted in the need for new and improved technology to test and substantiate the claims that are made about these garments.

Introducing the James Heal Performance Testing collection, a growing range of instruments designed for testing performance wear and technical textiles.

Our designers and technical specialists have created state of the art instrumentation for a range of applications including moisture management, dry rate testing, water repellency and wet abrasion. The collection includes improvements to existing methods for testing and game changing new technology to meet the requirements of an evolving industry.

The functionality of performance wear must be qualified by evidence, and our instruments have been designed to give brands, retailers and manufacturers confidence in their results.

1. ProDry
2. WickView
3. TruRain
4. AquAbrasion





ProDry

Quick-drying is an important attribute of textiles worn next to the skin, during sporting activities where the skin is likely to perspire. Without this characteristic, perspiration does not easily evaporate from the garment, which can leave a wearer feeling uncomfortable. ProDry replicates the conditions of use in order to verify that quick-drying fabrics are performing effectively.

This method is used in the development of quick-drying fabrics and garments such as: base layer garments, football kits, running gear, gym gear, socks and footwear fabrics.



Test Method

The instrument is capable of measuring the rate at which a textile specimen dries after being exposed to water. The specimen is placed on a heated plate, which is maintained at 37°C to simulate the temperature at which human skin begins to perspire. Water is dosed to the heated plate underneath the specimen to simulate perspiration.

The temperature drops when the specimen is wet and as the specimen dries, the temperature gradually increases. James Heal's TestWise software records this change in temperature and automatically calculates the drying time and rate of the tested specimen.

Why ProDry?

- Our internal testing found that increasing the number of data points used to calculate the test end time improved the accuracy and consistency of the results. ProDry records the temperature every second of the test and these points can be added into the graph once the test is complete.
- TestWise software can automatically set up the parameters of the test, find the end point of the test and calculate the results for the user. There is also a real time presentation of results on the test screen as the test takes place. There are two modes for water dosage to suit the user – automatic and manual.
- An adjustable temperature sensor ensures that ProDry complies with the AATCC standard regardless of how thick the fabric tested is. Setting gauges are provided for quick set up.
- Multiple fans, an air filter and positioning of air flow sensors provide consistent air flow which is measured and controlled by an integrated airflow sensor.

Standards

AATCC Test Method 201 Drying Rate of Fabrics: Heated Plate Method.



TestWise for ProDry



Home Screen

- 1 Real time presentation of results
Increased data points can be added to the final graph to improve the accuracy of the test
- 2 Visual warning are displayed when the Air Speed and Hotplate Temperature are out of tolerance
- 3 All test parameters visible and controlled from the home screen
- 4 TestWise software automatically finds the end point and stops the test
- 5 Test Status Bar
- 6 Settings menu to change language, units and display settings



Specimen Menu

- 1 Results can be displayed in a graph
- 2 Pinch and zoom functionality on graph, our most advanced TestWise Touch software
- 3 Up to six specimens displayed at the same time on the same graph
- 4 Option to focus on one specimen in more detail, and increase the data points used to plot slopes
- 5 Flexibility of determining results based on fabric behaviour
- 6 Test Start Time (average when multiple results are selected)
- 7 Test End Time (average when multiple results are selected)

WickView

Wicking is the transportation of liquid moisture within a textile by capillary action. It is an important characteristic of textiles which are likely to encounter liquid moisture during use. For sports and outdoor clothing, fabrics with good wicking ability can help to keep the wearer feeling cool and dry by effectively transporting perspiration away from the body to the outer surface of the garment where it can evaporate. WickView measures this movement of liquid moisture within a textile to give an accurate indication of its moisture management performance.

This method is beneficial in the development of wicking fabrics such as: Base layer garments, football kits, running gear, gym gear, socks and footwear fabrics.



Test Method

WickView is used to analyse the movement of moisture in multiple directions through a fabric.

A textile specimen is secured in place with a clamp ring so that the skin and face sides of the specimen are both exposed. Water is applied to the skin side of the fabric and spreads out as it is absorbed.

High definition cameras are positioned on either side of the specimen to record the movement of the water as it spreads out and through the fabric.

Images are processed and interpreted by James Heal's TestWise software, which provide useful information such as the Maximum Wetted Area, Maximum Wetted Distance and Spreading Speed.

The test chamber can be rotated in order to test a specimen in a vertical or horizontal orientation, providing more insightful results than just one orientation alone.

Why WickView?

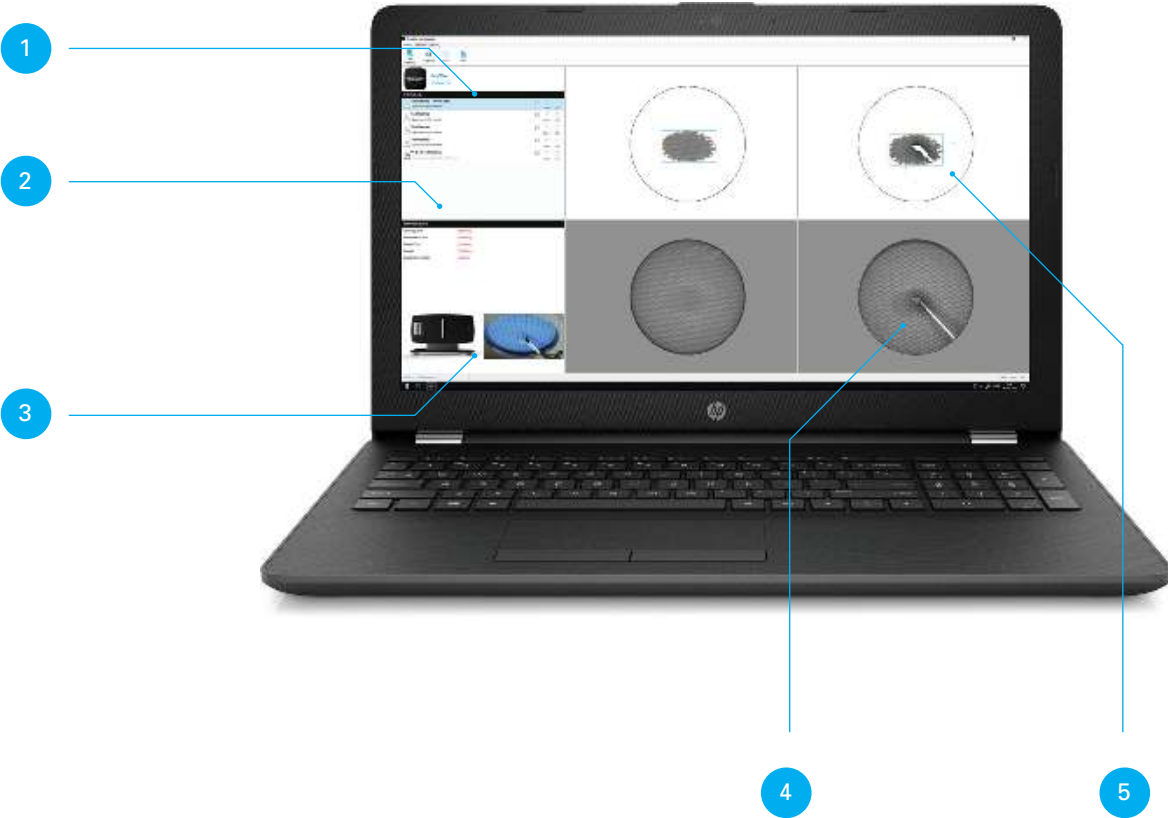
- WickView can be rotated to test specimens both vertically and horizontally, to better replicate how clothes are worn on the body, or how nonwovens are deployed in real-life usage
- The way WickView calculates records and calculates results is a truer representation of real life, in comparison to existing methods of moisture management testing. By examining the direction, shape, speed and actual wet area of a fabric, WickView is accurately measuring how sweat is drawn away from a wearer's body.
- TestWise for WickView shows the wicking process in detail, including live capture of the test, wet region boundary comparison and video playback of how the moisture spreads. The software can measure many different metrics, and results templates can be applied to show the user the information relevant to them. Results can be presented visually or in graph format.
- WickView is easy to clean and maintain, to ensure accurate testing every time.

Standards

WickView is new technology designed to improve how wicking is measured, a standard is being developed for this.

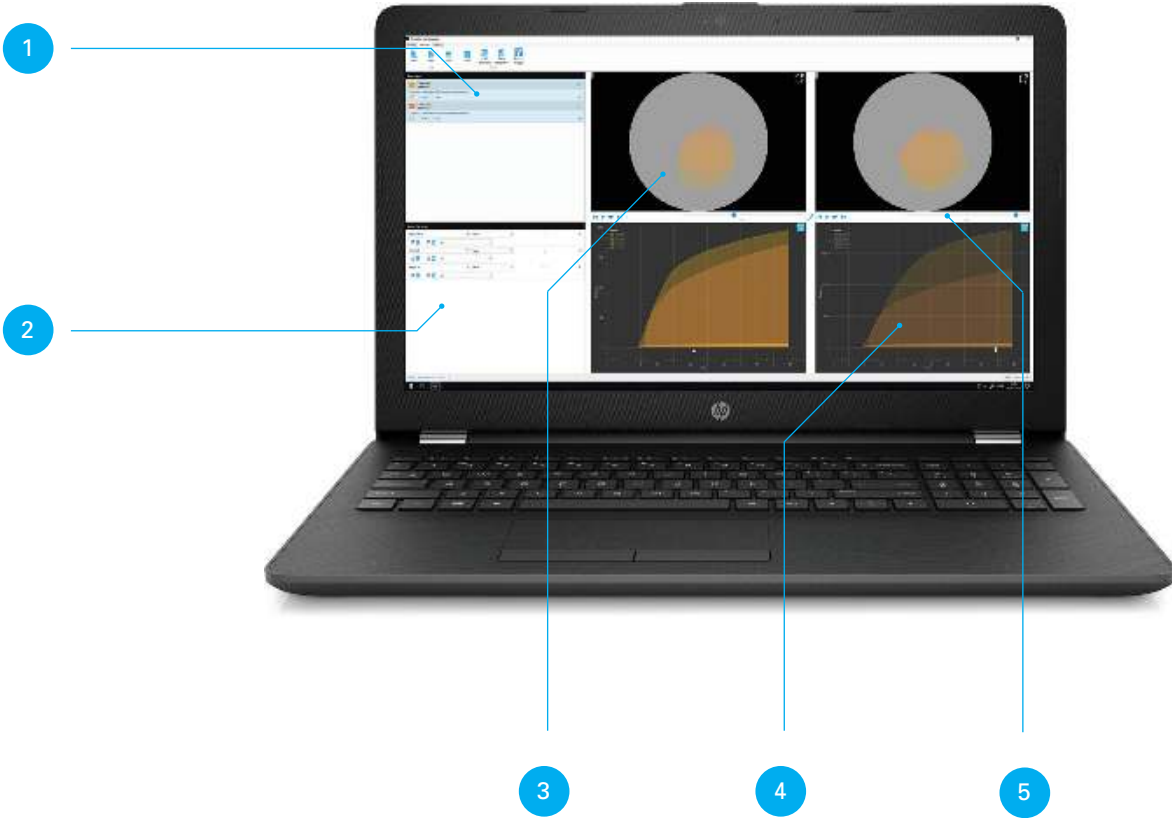


TestWise for WickView



Home Screen

- 1 Test methods are listed for automatic set up of test parameters
- 2 Instrument Status
- 3 The software shows and records which orientation the test is done in
- 4 TestWise automatically determines optimum light and exposure settings to capture images of the test, and a live preview is displayed here
- 5 Wet Region Preview



Results Screen

- 1 Specimen Result Files
- 2 Toggle buttons can be used to display different data on the graphs
- 3 A range of metrics can be calculated by TestWise, including Wetted Area, Maximum Wetted Area, Maximum Wetted Distance, Spreading Speed and Directional Spreading Speed
- 4 Multiple test specimens can be compared, including saved results from previous tests
- 5 Playback Control Bar



TruRain

Water repellency is an important function of technical fabrics, which are intended to protect the user in wet and rainy conditions. The TruRain water repellency tester replicates the conditions of use and therefore provides an accurate indication of fabric performance.

This method is used in the development of water repellent garments and accessories such as outdoor jackets, over trousers, walking boots, backpacks, tents and awnings.



Test Method

TruRain is an artificial rain shower testing instrument for determining the water repellency of textiles. It measures three aspects of water repellency in a single test:

- 1 Surface water repellency is assessed by visually comparing the tested specimen with a photographic grading scale (similar to the spray rating tester).
- 2 Resistance to water absorption is evaluated by weighing the specimen before and after test. The after test measurement is taken after the specimen has been centrifuged to remove any excess surface water.
- 3 Resistance to water penetration is determined by collecting and recording any water which passes through the test specimen into the specimen holder cups.

Why TruRain?

- The definitive Bundesmann tester, designed with precision in mind to reduce inconsistency and improve the accuracy and repeatability of results
- An integrated, automated shower guard diverts the water flow before and after each test to keep the operator dry when inserting and removing the specimens. Water can be switched on and off with one button, and a water pressure regulator guarantees consistency wherever in the world the test takes place.
- Our TestWise software is user friendly, automating the process of testing and alerting the user of when to grade the specimens based on the ISO standard
- The 304 evenly distributed nozzles which produce the rain drops for the test are easy to clean and maintain.
- The cups can be easily removed individually using a twist motion, minimising the amount of space required for set up. All four cups can be removed together on their carousel.

Standards

ISO 9865 – Determination of water repellency of fabrics by the Bundesmann rain-shower test



AquAbrasion

Textile abrasion is the surface wear caused by rubbing with another material, this can often be against itself when a garment is on the body. The consumer expects textiles and other products to withstand a certain amount of deterioration, which is tested in dry conditions by the Martindale instrument.

The market for outdoor wear, waterproofs and garments designed to perform while damp or wet is growing, and current testing does not take into account the wet conditions these garments will abrade in. Our tests show that fabrics subjected to wet abrasive stresses degenerate at a faster rate both aesthetically and physically, and AquAbrasion accurately represents these conditions.

This method is beneficial for development, and proving performance, of raincoats, cagoules, waterproof trousers, rucksacks and footwear.



Test Method

AquAbrasion is a Wet Abrasion Tester based on the traditional Martindale instrument. It uses a controlled pump system to dose fabric specimens with liquid which keeps the specimen wet for the duration of the test. Drains are recessed into the main table to prevent overspill when conducting testing.

Deionised water can be used to replicate rain, or a perspiration solution can be used to replicate sweat. Chlorinated water and saliva solutions are suitable for use on this instrument.

AquAbrasion can also conduct the full range of dry Martindale testing,

Why AquAbrasion?

- Moisture is controlled and delivered in a standard dosage, which prevents human error and ensures repeatability and accuracy of results.
- AquAbrasion abrades fabric against itself while wet to replicate real life situations, such as a sleeve on a cagoule rubbing against the body of a wearer. The testing replicates the use of outdoor wear in inclement conditions, providing a more representative test for brands and manufacturers.
- The instrument can be set up to perform wet and dry abrasion concurrently on different stations, and can also test different liquid solutions concurrently using multiple beakers. AquAbrasion retains the full functionality of a Martindale 9 station instrument.
- The TestWise Touch software is easy to use by operators of all skill levels.

Standards

ISO 17704, ISO 20344 Part 6.12 for wet testing. For dry testing, please visit www.james-heal.co.uk/martindale for a full list of standards.







James Heal, Richmond Works
Halifax, HX3 6EP, England

Tel +44 (0) 1422 366355

Fax +44 (0) 1422 352440

info@james-heal.co.uk

www.james-heal.co.uk